

CLEAR

Clean, Leakage-free Evaluation And Reporting
an honest evaluation harness for highway trajectory prediction

1 INPUTS

- **A dataset**
highD · exiD · NGSIM
(you provide it — CLEAR ships none)
- **Your model**
subclass Predictor
predict(ws) → (N, T, 2)

2 CLEAR PIPELINE

- **Canonical frame**
per-window, handles ramps
- **Leakage-safe split**
+ PASS / WARN verdict
- **CV · CA · IDM**
honest baselines
- **Metrics**
FDE/RMSE · CIs · CA win-rate
strata · best-of-k · collisions

3 EVAL CARD

- **Protocol**
+ leakage verdict
- **Accuracy ladder**
every model vs CA
- **Stratified error**
cruise / mild / hard
- **Admissibility**
predicted collisions
- **5-point checklist**
JSON · Markdown · figure

WHY IT MATTERS

- **Leakage inflates.**
Sliding windows share vehicles and recordings across folds; CLEAR forbids it by default.
- **Averaging hides the tail.**
~77% of windows are trivial cruising; one RMSE buries the safety-critical cases.
- **CA is a hard baseline.**
A zero-parameter constant-acceleration model is unbeaten on clean highway data.
- **Accuracy ≠ safety.**
The lowest-error model can forecast the most rear-end collisions — CLEAR measures both.